

CHECKLIST
SMALL CRAFT - Electric propulsion systems
Ref.: ISO 16315:2026 [ISO 16315:2026]
Not yet harmonised
 Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	



Installation documentation as per Annex B			
Subject to check	Clause	Requirements	Checked ?
Scope: the installed electrical propulsion operates in the following ranges either individually or in combination: - direct current of less than 1 500 V DC; - single-phase alternating current up to AC 1 000 V; - three-phase alternating current up to AC 1 000 V.		[Yes ?]	
Type and arrangement of electrical propulsion system to be indicated as specified in the wire diagram and additional technical documentation			
DC: sourced from battery(s) or DC generator(s) or AC/DC converters from an AC source - DC fully insulated two-wire system for any system voltage - DC two-wire system with negative earth/bonded for systems below a maximum voltage of 60 - DC an earth/bonded three-wire system AC: sourced from alternator(s) or a DC/AC converter from a DC source [e.g. battery(s)] - AC single-phase two-wire insulated (IT) - AC single-phase two-wire with earthed neutral (TT or TN-C without hull return), or system having - AC single-phase three-wire with mid-point earthed, both neutral and protective conductor (PE) - AC three-phase three-wire insulated (IT) - AC three-phase four-wire with earthed neutral (TT or TN-C without hull return), or TT (when - AC three-phase five-wire with mid-point earthed, both neutral and protective conductor (PE) earthed at the energy source without hull return (TN-S).			

Watercraft manufacturer:	
Watercraft model name:	



Environmental factors	4.5	
The propulsion system and all its components shall be designed for safe operation in the environment in which it is intended to be used.	4.5.1	
In case that the system compensates for high or low temperatures by delivering reduced performance, this reduction does not cause sudden or abrupt changes in speed or direction. Any reduced performance due to environmental conditions is indicated to the operator.	4.5.1	
To achieve and maintain the ambient temperature at or below the maximum at which the equipment is designed to operate adequate ventilation is provided. Enclosures for electrical equipment are mechanically strong and rigid, and mounted so that the equipment will not be affected by the distortion, vibration or movement of the craft's structure that occur during normal operation of the small craft.	4.5.2	
Natural air-cooling of enclosures shall be designed with sufficient ventilation openings or with sufficient cooling surface to dissipate the heat so that enclosed equipment will operate within their design temperature limits.	4.5.3	
Electrical ratings	4.6	
Electrical and electronic equipment shall function under the maximum voltage and frequency variations and harmonic distortion which can occur in the electric propulsion system in normal operation.		
The basic limits for DC and AC have been taken into account. AC variable frequency drives conform with the manufacturers design parameters.		
Equipment	4.7	
Transformers conform with IEC 61558-2-4 and are installed in well-ventilated locations.	4.7.1	
Semiconductor converters conform to the requirements in Table 13 IEC 60146-1-1:2024 and are installed according to the component manufacturers instructions.	4.7.2	
Motors meet the requirements of IEC 60034-1 and the electric propulsion systems shall be rated according to ISO 8665-2.	4.7.3	
Enclosures	4.8	
Energized parts of electrical equipment are guarded against accidental contact by the use of enclosures.	4.8.1	
Access to enclosures housing energized parts of the electrical system require the use of hand tools or have a protection of at least IP2X.	4.8.2	

Watercraft manufacturer:	
Watercraft model name:	


Identification of equipment and conductors **4.9**

All electrical equipment shall be marked with the following: 4.9.1

All electrical enclosures shall be marked with the following:

a) the electrical rating in volts and amperes or volts and watts; 4.9.2

b) the phase and frequency, if applicable.

All cables and conductors are permanently identified. 4.9.3

Insulation colours used conform to IEC 60445 and for active (phase) or neutral AC conductors, yellow, green or green with a yellow stripe insulation are not used. 4.9.3

Active (phase) and neutral conductors of an AC system are identified by insulation colour, by numbering or by other means on the wiring diagram for the system supplied with the craft. 4.9.3

All DC negative conductors are identified by black or yellow insulation.

Craft equipped with an AC electrical system which uses black insulation for live conductors, yellow insulation is used for DC negative conductors of the DC system. 4.9.3

Black or yellow insulation are not be used for DC positive conductors.

Means of identification other than insulation colour for DC negative conductors if the craft is fitted with multiple independent DC systems is permitted if permanently marked on the conductor and craft wiring diagrams of the electrical system(s). 4.9.3

All conductors greater than DC 60 V maximum voltage indicate the correct polarity and be orange in colour or permanently marked as exceeding DC 60 V maximum voltage. 4.9.3

All conductors for systems operating at a maximum voltage greater than AC 250 V are identified as such. 4.9.3

Steering and throttle controls **4.10**

Electrical/electronic control systems for steering, ahead/astern, and speed or power meet the requirements of ISO 25197.

Watercraft manufacturer:	
Watercraft model name:	

**Electrical equipment in the vicinity of batteries****4.12**

Ignition protection is not required if the battery, or battery bank is separated in a sealed and ventilated box, where the ventilation opening is located at the top and the hose leads directly to the atmosphere.

All electrical equipment located in the same vicinity as a battery, that has the possibility of emitting flammable gas or substances during normal operation shall be ignition protected in accordance with ISO 8846 or IEC 60079-7.

Hazardous areas**4.13**

Areas and equipment that can endanger the safety of the crew (e.g. battery compartments, electrical assemblies) are:

- a) accessible only with the use of tools or keys;
- b) labelled appropriately to the hazard they present by the use of the symbols in Figure 1, as applicable.

Fire Protection**4.14**

Fire protection equipment shall be specified by the propulsion system manufacturer. Hereby, the equipment may depend on component types, energy storage systems, and location of components.

Controls, monitoring, system alerts and trips alarms**5**

Controls are implemented and installed to control, and also limit, energy flow to prevent damage between the battery and/or power sources and the rest of the propulsion system in either direction.

5.1.1

For each propulsion motor/propeller system the following controls and switches are included:

- On/Off for each system
- Ahead/Neutral/Astern
- Propeller speed control

5.1.1

A readily accessible means of emergency stop is installed at all control stations that disables the propulsive thrust, while maintaining all other systems under operation to prevent propeller injury.

5.1.2

If activated, the propulsion shall be reactivated using the fault trip reset; see cl. 5.1.3

A fault trip reset is provided for the propulsion system being effective in one of the following conditions:

- a) no propulsion system trip is present;
- b) control for ahead/neutral/astern control is set at neutral position;
- c) propeller speed control set at minimum speed;
- d) the "emergency stop" (see 5.1.2) and/or a "battery disconnect" is not operated.

5.1.3

Watercraft manufacturer:	
Watercraft model name:	



Instruments, alerts and trip alarms	5.2	
Operating mode and status is provided for each electric propulsion system and installed at the main steering control station.		
On craft with multiple control stations, each station other than the main provides an indicator of system alerts and fault trip alarms.	5.2.1	
System alerts are provided as applicable.	5.2.2	
Fault trip alarms are provided as applicable.	5.2.3	
Protection against electrical shock	6.	
Protection against direct accidental contact is provided.	6.1.1	
Exposed non-current-carrying conductive parts are connected to earth or to the negative pole or the earthed conductor.	6.1.2	
The system consisting of all components has sufficient insulation resistance more than 500 Ω/V for DC systems and 1000 Ω/V for AC systems.	6.1.3	
Earthed two-wire DC systems and earthed neutral AC systems have an automatic disconnection of supply to the electric propulsion system under fault-to-earth conditions.	6.2	
Fault-to-earth monitoring is provided for DC fully insulated systems and DC 3-wire systems.	6.3	
Fault-to-earth tripping in AC non-neutral earthed systems (IT-type system) is provided.	6.4	
Protection against overcurrent	7.	
Each propulsion system is protected against overload by an overcurrent protection device such as circuit-breaker(s) or fuse(s) whereas circuit-breakers are selected in accordance with IEC 60947-1 or IEC 60898-1.	7.1	
Each fuse or circuit-breaker at the source of a circuit shall be designed to protect the conductor with the smallest cross-section in the circuit being protected.		
The characteristics of protective devices is selected to comply with this section.	7.2	
The outgoing circuit(s) from a battery are protected by corresponding overcurrent devices withing the required distance.	7.3	
Overcurrent protection devices in fully insulated systems are provided to comply with this section.	7.4	
Battery monitoring and installation	8.	
The general arrangements are met as required with this section.	8.1	
Isolation of batteries or battery banks is implemented as required with this section.	8.2	
Operational switching of batteries or battery banks is arranged to meet this section.	8.3	
Permanently energized circuits are installed as described in this section.	8.4	
Ventilation is provided in accordance with the battery manufacturers recommendations.	8.5	

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
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Electrical installations	9.	
The general arrangements of the electrical installations are met as required with this section.	9.1	
The segregation of electrical propulsion system cables are separated as required.	9.2	
Testing	10.	
The entire electrical propulsion system has been tested after completion taking into account:		
- Earthing and bonding		
- Insulation resistance		
- The electrical/electronic control systems		
- On load near its normal working load		
- Voltage drop		

Installation documentation as per Annex B			
Subject to check	Clause	Requirements	Checked ?
1 For the battery group data including the following: - battery technology/type; - prospective short circuit current in (kA) and time constant in (s); - ventilation and temperature parameters; -Ampere hours (Ah).	Annex B	[Yes ?]	
2 Generator data including the following: - power (kW), rated voltage, power factor, frequency, other; - necessary reactance's and time constants; - maximum operating temperature.	Annex B	[Yes / NA ?]	
3 Propulsion motor data including the following: - power (kW), rated voltage, power factor, frequency, other; - regeneration performances; - braking data; - maximum operating temperature.	Annex B	[Yes ?]	
4 Charger data including the following: - power (kW), rated voltage, power factor, frequency, other; - short-circuit current (kA); - maximum operating temperature.	Annex B	[Yes ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

Checklist_Evaluation_Module B_G en260703

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Further documentation to be provided to IMCI

Overcurrent protection for each propulsion unit: position and rating		[Yes ?]	
Wire diagram including conductor length and type		[Yes ?]	
Voltage drop calculation	10.6	[Yes ?]	
Proof of separation of the DC system for propulsion and household [see RCD, Annex I, para 5.3]		[Yes ?]	
Proof that transformer comply with the applicable IEC standard.	4.7.1	[Yes ?]	
Proof that semiconductor converters comply with Table 13 IEC 60146-1-1:2024	4.7.2	[Yes ?]	
Proof that the electrical motor complies with IEC 60034-1	4.7.2	[Yes ?]	
Enclosure positioning and rating	4.8	[Yes ?]	
Information about the identification of all electrical equipment and enclosures	4.9	[Yes ?]	
Proof of compliance for an electric/electronic throttle according to EN ISO 25197	4.10	[Yes / NA ?]	
Electrical equipment in the vicinity of batteries	4.12	[Yes ?]	
Hazardous area protection	4.13	[Yes ?]	
Fire protection	4.14	[Yes ?]	
Information and proof about fault tripping of controls, its reset and emergency stop and manual emergency control mode	5.1	[Yes ?]	
Information and proof for instruments, alerts and trip alarm	5.2	[Yes ?]	
Information and proof for protection against electrical shock: accidental contact and insulation resistance	6.1	[Yes ?]	
Information and proof for fault to earth automatic disconnection	6.2, 6.3, 6.4	[Yes ?]	
Information and proof of the overcurrent protection, type and characteristic.		[Yes ?]	
Information about the battery monitoring and installation	8	[Yes ?]	
Documentation about the compliance of the electrical installation which refers to IEC 60092-352 and other standards	9	[Yes ?]	
Earthing and bonding testing	10.2	[Yes ?]	
Insulation resistance of all components in the system.	10.3	[Yes ?]	
Electric/Electronic control systems for propulsion motor	10.4	[Yes ?]	
On load test and inspection of electrical propulsion systems, and associated switch gear and control gear	10.5	[Yes ?]	
Voltage drop;	10.6	[Yes ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

Checklist_Evaluation_Module B_G en260703

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Instructions/Warnings to be included in the owner's manual			
Subject to check	Clause	Requirements	Checked ?
Block diagram of the propulsion system and a description of the installation location of the electric assemblies.	A.1	[Yes ?]	
Diagram(s) identifying the electrical circuits of the craft with the locations of electrical devices in the craft and identification of conductors by colour or other means;	A.1	[Yes ?]	
Location and description of functions of electrical controls, dials, switches, fuses and also circuitbreakers installed on the panel board;	A.1	[Yes ?]	
Location and description of functions of electrical controls, dials, switches and fuses, as well as circuit-breakers installed on the panel board.			
Instructions on how to leave the crafts electrical propulsion system when unattended;	A.1	[Yes ?]	
Description of display information as required by Clause 5;	A.1	[Yes ?]	
Marine installation, alterations and maintenance must be performed by a competent marine electrical technician.	A.1	[Yes ?]	
Manufacturer instructions for any electrical machinery provided for consumer use	A.1	[Yes ?]	
The charging procedure of the system shall be described. If different charging options are available for the system, each one shall be described. Applicable warnings with respect of the hazard of fire and risk of burns shall be stated.	A.1	[Yes ?]	
Applicable notice and warnings shall be described if the system is installed in a craft which is designated for trailering.	A.1	[Yes ?]	
Operational temperature limits of the propulsion system			
Instructions on appropriate actions for reacting to fires in the craft electrical propulsion system.	A.1	[Yes ?]	
Following advisory instructions are provided for the owner's manual to never: - work on the electrical installation while the system is energised; - modify the propulsion system, battery type and system components; - alter or modify the rated current amperage of overcurrent protective devices; - install or replace electrical appliances or devices with components which exceed the rated current amperage of the circuit.	A.2	[Yes ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X